

1. Administrative & Study Identification

**Full Study Title:** An Extension Study of Subcutaneous Secukinumab in Patients With Juvenile Psoriatic Arthritis (JPsA) and Enthesitis Related Arthritis (ERA) **Short Title/Acronym:** Secukinumab JPsA/ERA Extension **Protocol Number:** CAIN457F2304E1 **NCT Number:** NCT03769168 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Secukinumab (Cosentyx) **ATC Code:** L04AC10 **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To gather comprehensive data and evaluate the long-term efficacy, safety, and tolerability of subcutaneous secukinumab up to 4 years (up to Week 312) in patients with Juvenile Idiopathic Arthritis (JIA) subtypes of Juvenile Psoriatic Arthritis (JPsA) and Enthesitis Related Arthritis (ERA). **Secondary Objectives:** To evaluate JIA ACR 50/70/90/100 response over time, changes in each JIA ACR Core Component, and Inactive Disease status over time up to Week 312.

**2.2 Background & Rationale** To address the need for long-term efficacy and safety data of interleukin-17A (IL-17A) inhibition in pediatric patients. Participants who demonstrated clinical benefit and successfully completed the 104-week core study (NCT03031782 / CAIN457F2304) require an extension study to evaluate the durability of response, safety profile, and optimized weight-based dose escalation of secukinumab over an extended period.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional (Extension study) **Trial Status:** COMPLETED **Control Type:** Single-arm (Open-label roll-over extension) **Blinding:** Open-label **Randomization:** N/A (All participants continue on active treatment)

**3.2 Planned Enrollment & Duration Target \$N\$:** 55 patients **Number of Sites:** 23 locations **Estimated Study Start:** 07-Jun-2019 **Primary Completion:** 07-Nov-2024 (Completed)

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Patients had to have participated in the core study CAIN457F2304 and completed the entire treatment period up to and including Week 104. 2. Patients had to be deemed by the investigator to benefit from continued secukinumab therapy. 3. Age 2 years and older at core study enrollment.

**4.2 Exclusion Criteria** 1. Patients with plans for administration of live vaccines during the extension study period. 2. Patients taking any other concomitant biologic immunomodulating agent(s) except secukinumab. 3. Patients who were deemed not to be benefiting from the study treatment based on a lack of improvement or worsening of their symptoms.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Secukinumab 75 mg or 150 mg administered subcutaneously every 4 weeks (based on body weight  $<50$  kg or  $\geq 50$  kg). For patients whose signs and symptoms were not fully controlled, the dose could be escalated (e.g., from 75 mg to 150 mg, or 150 mg to 300 mg for patients  $\geq 50$  kg in a two-step escalation). **Route of Administration:** Subcutaneous (SC) injection via pre-filled syringes (PFS). **Duration of Treatment:** Up to 4 years of active treatment in the extension phase (from Week 104 to Week 312 overall).

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard non-biologic DMARDs (e.g., methotrexate) and NSAIDs according to standard of care. **Prohibited:** Administration of live vaccines and concurrent use of any other biologic immunomodulating agents.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days)      | Key Activities/Procedures                                                                                                              |
|----------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Screening/Baseline | Week 104   Rollover from core study CAIN457F2304; Baseline efficacy & safety assessment                                                |
| 2        | Interim            | Every 4 to 12 weeks   SC Dosing, JIA ACR assessments, CHAQ, Safety Labs (e.g., Weeks 116, 128, 140, 156, 180, 208, 232, 260, 284, 308) |
| 3        | End of Study       | Week 312   Final efficacy (JIA ACR30) and safety evaluations, Exit procedures                                                          |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Clinical response to treatment according to JIA ACR30 improvement in disease activity over time up to Week 312. **Measurement Method:** JIA ACR response criteria consisting of 6 core criteria: Physician global assessment VAS (0-100), Parent/patient global assessment VAS (0-100), Functional

ability via CHAQ, Count of active joints, and Index of inflammation via CRP concentration.

**7.2 Secondary & Exploratory Endpoints** 1. JIA ACR 50/70/90/100 response over time up to Week 312. 2. Change from baseline for each JIA ACR Core Component and Inactive Disease status over time up to Week 312.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard longitudinal collection via routine clinical visits evaluating long-term safety, tolerability, and immunogenicity up to 4 years. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor safety desk. **Data Monitoring Committee (DMC):** Independent safety monitoring board overseeing the long-term pediatric safety profiles.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Full Analysis Set (FAS) for efficacy and Safety Set for adverse events. **Sample Size Justification:** Determined strictly by the number of eligible patients rolling over from the core study CAIN457F2304 (\$N=55\$); no formal power calculation was required for the extension. **Handling of Missing Data:** Descriptive statistics over time; no formal statistical hypothesis testing was planned for the primary endpoint as per the SAP.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and safety tracking. **Audit Trail:** Validated eCRF data capture with comprehensive audit trails and data clarification forms.

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## 11. Study Identifiers & Governance

**Primary ID:** CAIN457F2304E1 **Registry Number:** NCT03769168 **Posting ID:** 627873271801324466 **Sponsor/Lead Organization (Company):** Novartis Pharmaceuticals **EMA URL:** [https://www.ema.europa.eu/en/documents/product-information/cosentyx-epar-product-information\\_en.pdf](https://www.ema.europa.eu/en/documents/product-information/cosentyx-epar-product-information_en.pdf) **Study Title:** An Extension Study of Subcutaneous Secukinumab in Patients With Juvenile Psoriatic Arthritis (JPSA) and Enthesitis Related Arthritis (ERA) **Official Regulatory Title:** An extension study of subcutaneous secukinumab

to evaluate the long-term efficacy, safety and tolerability up to 4 years in patients with Juvenile Idiopathic Arthritis subtypes of Juvenile Psoriatic Arthritis and Enthesitis Related Arthritis

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## 12. Clinical Framework (JPsa & ERA Specific)

**Disease Areas:** Juvenile Psoriatic Arthritis, Enthesitis Related Arthritis **Preferred / Condition Name:** Arthritis **Preferred UMLS Name:** Juvenile Idiopathic Arthritis **Definition:** Acute or chronic inflammation of JOINTS. **Semantic Type:** Disease or Syndrome **MeSH Code:** D001168 **HPO Code:** HP:0001369 **SNOMED ID:** 3723001 **Diagnosis Threshold:** Completion of the 104-week core study with demonstrated clinical benefit **Medical Methodology:** Interleukin-17A (IL-17A) inhibitor therapy **Optimization Target:** Maintenance of JIA ACR30 response or higher, and dose escalation for symptom control

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## 13. Study Procedures & Methodology

**Management Pathway:** Long-term open-label biologic therapy extension **Education Protocol (HCP):** Protocol-specific dosing, body-weight stratification ( $<50$  kg or  $\geq 50$  kg), and dose-escalation guidelines **Education Protocol (Patient):** Subcutaneous injection training for pre-filled syringes (PFS) **Quality Audit Mechanism:** Routine clinical monitoring and eCRF data validation

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Up to 208 weeks (4 years) in the extension phase (Total 312 weeks) **Onsite Visit Cadence:** Every 4 to 12 weeks for clinical assessments (e.g., Weeks 116, 128, 140, 156, 180, 208, 232, 260, 284, 308, 312) **Remote Monitoring Frequency:** Standard ongoing pharmacovigilance **Checklist Review Interval:** Every 4 weeks (coinciding with the 4-week dosing schedule)

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|------|------|------|------|
| Role                     | Name   | Signature | Date          | :--- | :--- | :--- | :--- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |